

- 10** A group of researchers brought 180 koalas to an island and tracked the number of koalas over a few years.

x (years)	0	0.5	1	2	3	4	5	6	6.5	7	7.5
y (no of animals)	180	177	175	172	p	172	175	180	183	187	191

- (a) Using a scale of 2 cm on both axes to represent 1 year on the x -axis and 2 animals on the y -axis for $170 \leq y \leq 192$, plot the values given and join them with a smooth curve.

[3]

- (b) Use your curve to estimate the value of p , corrected to nearest integer.

Answer _____ [1]

- (c) By drawing a tangent, find the gradient at $x = 6$.

Answer _____ [2]

- (d) The number of koalas (y) turns out to be related to an unknown quadratic function $y = (x + a)^2 + b$. Find the value of a and of b .

Answer $a =$ _____ $b =$ _____ [2]

- (e) The researchers also brought 180 penguins to the same island and noticed that the number of penguins increased at a steady rate of 1 per year.
- (i) Write down an equation connecting the number of penguins, y , and the number of years, x .

Answer _____ [1]

- (ii) On the same graph in (a), draw the graph of the equation in (e)(i) and state the point of intersection (x , y).

Answer _____ [2]

- (iii) What would this point of intersection represent in this research?

Answer

[1]

- 11 Charlotte takes the taxi every day. She is thinking of buying a car and wants to pay off the car loan as soon as possible. The tables below give the information Charlotte can use.

Charlotte's financial situation	
Money set aside to buy a car	\$50000
Monthly salary	\$4500

Information on the car and car loan	
Car Price	\$96000 Customer must take a minimum loan of 50% from the car dealer.
Loan Period (Number of years, x)	$2 \leq x \leq 7$, x is an integer - Customer needs to decide on the loan period on the day he or she buys the car.
Interest rate of loan	2.45% per annum simple interest

